Project Executable Files

Date	2 July 2026
Team ID	xxxxxx
Project Name	Smart Lender
Maximum Marks	3 Marks

Project Executable Files

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	NA
5	Environment configuration file (.env.example or similar)	NA

6	Deployed application URL (if hosted)	No (Local Deployment)
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

(Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

```
Smart_Lender/
│
├── Dataset/
│   └── loan_prediction.csv
│
├── Training/
│   └── Smart Lender Using ML.ipynb
│
├── Flask/
│   ├── app.py
│   ├── requirements.txt
│   ├── xgb_model.pkl
│   ├── scale1.pkl
│   │
│   ├── Templates/
│   │   ├── index.html
│   │   └── result.html
```

```
|    |
|    └──── Static/
|    ├──── style.css
|    └──── background.jpeg
|
├──── Project_Screenshot_Images.pdf
|
└───── README.md
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	Not Hosted (Runs Only Local Flask Application)
Login Credentials (Demo)	Not Applicable (No Login Required)
Platform / Hosting Provider	Local Flask Development Server
Repository Link	https://github.com/sivasankarpilli/smart-lender-loan-prediction
Demo Video Link	

Step 4: Run Instructions

Pre-requisites:

Hardware Requirements:

-
- Processor: Intel i3 or above
- RAM: 4 GB (8 GB recommended)
- Storage: 10 GB free space
- System Type: 64-bit OS
- Display: 1366×768 or higher

Software Requirements:

- Windows 10/11 / Linux / macOS: Operating System
- Anaconda Navigator: Python environment management
- Python 3.8+: Core programming language
- PyCharm / VS Code: Code editor
- Google Chrome / Edge: Web browser for Flask app
- Flask
- XGBoost

Installation Steps:

Step 1: Download the Project
Download or clone the project repository.
git clone <repository-url> cd Smart-Lender
Alternatively, extract the provided ZIP file into a folder.

Step 2: Create a Virtual Environment
Create a virtual environment to isolate project dependencies.
Windows:
python -m venv venv
venv\Scripts\activate
Linux/macOS: python3 -m
venv venv source
venv/bin/activate

Step 3: Install Required Packages Install all
required Python libraries.
pip install flask pip
install pandas pip
install numpy pip install
scikit-learn pip install
xgboost pip install
joblib pip install
matplotlib pip install
jupyter
Or, if a requirements.txt file is available: pip install
-r requirements.txt

Step 5: Run the Flask Application
Navigate to the Flask directory and start the server.
cd Flask python app.py
The application server will start successfully.
Example output:
* Running on http://127.0.0.1:5000/

Step 6: Open the Application Open any
web browser and visit:
http://127.0.0.1:5000
The Smart Lender web application homepage will appear.

Step 7: Test Loan Prediction

1. Enter applicant details in the form.
2. Click the Predict button.
3. The system processes the data using the trained XGBoost model.
4. The prediction result (Loan Approved/Rejected) is displayed on the result page.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	Model prediction depends on training dataset quality	Can be improved by training with a larger and more diverse dataset
2	Application runs locally by default	Can be deployed on IBM Cloud, Render, or another cloud platform
3	No User Authentication	Can be release in future version.